

UAV Slung-Payload: Kinematic Observations

Controlled Figure-8 Speed Sweep - 2026-07-22

Purpose

This experiment checks whether the tuned geometric controller advantage remains visible when the Figure-8 angular rate changes.

Setup

- Vehicle: `iris_depth_payload`
- Payload: native ball-joint slung payload
- Controllers: PX4 position/velocity baseline and tuned geometric attitude/thrust controller
- Angular rates: `0.15`, `0.20`, `0.25`, `0.30` rad/s
- Trials per speed and controller: `3`
- Flight duration per trial: `75.0` s
- Geometric hover thrust: `0.72`
- Validation gates: tracking sample count, PX4 local-position magnitude, steady altitude, and payload swing telemetry
- Raw per-trial telemetry CSV retention: `False`

Controller Improvement

omega_rad_s	metric	baseline_mean	geometric_mean	improvement
0.1500	mean_3d_error_m	0.1788	0.2202	-23.1138

omega_rad_s	metric	baseline_mean	geometric_mean	improvement
0.1500	rms_3d_error_m	0.1910	0.2292	-20.0196
0.1500	mean_lateral_swing_m	0.3546	0.3557	-0.3070
0.1500	mean_cable_angle_deg	20.8413	20.9007	-0.2849
0.2000	mean_3d_error_m	0.2827	0.3127	-10.6397
0.2000	rms_3d_error_m	0.3006	0.3247	-8.0072
0.2000	mean_lateral_swing_m	0.4634	0.4393	5.1904
0.2000	mean_cable_angle_deg	27.7412	26.1677	5.6721
0.2500	mean_3d_error_m	0.4459	0.3387	24.0418
0.2500	rms_3d_error_m	0.4759	0.3543	25.5636
0.2500	mean_lateral_swing_m	0.5642	0.5147	8.7694
0.2500	mean_cable_angle_deg	34.5840	31.1520	9.9238
0.3000	mean_3d_error_m	0.6760	0.3841	43.1727
0.3000	rms_3d_error_m	0.7194	0.4006	44.3094
0.3000	mean_lateral_swing_m	0.6569	0.5853	10.8903
0.3000	mean_cable_angle_deg	41.4097	36.0678	12.9001

Profile Metrics

omega_rad_s	profile	metric	mean	std	min	max
0.1500	baseline	mean_3d_error_m	0.1788	0.0005	0.1785	0.1794
0.1500	baseline	rms_3d_error_m	0.1910	0.0004	0.1907	0.1914
0.1500	baseline	mean_lateral_swing_m	0.3546	0.0000	0.3546	0.3546

omega_rad_s	profile	metric	mean	std	min	max
0.1500	baseline	mean_cable_angle_deg	20.8413	0.0013	20.8401	20.8426
0.1500	geometric	mean_3d_error_m	0.2202	0.0004	0.2199	0.2206
0.1500	geometric	rms_3d_error_m	0.2292	0.0003	0.2289	0.2295
0.1500	geometric	mean_lateral_swing_m	0.3557	0.0002	0.3555	0.3559
0.1500	geometric	mean_cable_angle_deg	20.9007	0.0090	20.8925	20.9103
0.2000	baseline	mean_3d_error_m	0.2827	0.0007	0.2820	0.2834
0.2000	baseline	rms_3d_error_m	0.3006	0.0009	0.2998	0.3016
0.2000	baseline	mean_lateral_swing_m	0.4634	0.0005	0.4627	0.4637
0.2000	baseline	mean_cable_angle_deg	27.7412	0.0343	27.7016	27.7622
0.2000	geometric	mean_3d_error_m	0.3127	0.0008	0.3120	0.3136
0.2000	geometric	rms_3d_error_m	0.3247	0.0005	0.3242	0.3252
0.2000	geometric	mean_lateral_swing_m	0.4393	0.0007	0.4386	0.4401
0.2000	geometric	mean_cable_angle_deg	26.1677	0.0466	26.1198	26.2128
0.2500	baseline	mean_3d_error_m	0.4459	0.0006	0.4453	0.4465
0.2500	baseline	rms_3d_error_m	0.4759	0.0005	0.4755	0.4765
0.2500	baseline	mean_lateral_swing_m	0.5642	0.0003	0.5639	0.5644
0.2500	baseline	mean_cable_angle_deg	34.5840	0.0153	34.5675	34.5977
0.2500	geometric	mean_3d_error_m	0.3387	0.0013	0.3372	0.3395
0.2500	geometric	rms_3d_error_m	0.3543	0.0012	0.3529	0.3549
0.2500	geometric	mean_lateral_swing_m	0.5147	0.0002	0.5145	0.5150
0.2500	geometric	mean_cable_angle_deg	31.1520	0.0163	31.1408	31.1707

omega_rad_s	profile	metric	mean	std	min	max
0.3000	baseline	mean_3d_error_m	0.6760	0.0039	0.6716	0.6791
0.3000	baseline	rms_3d_error_m	0.7194	0.0033	0.7158	0.7223
0.3000	baseline	mean_lateral_swing_m	0.6569	0.0006	0.6564	0.6575
0.3000	baseline	mean_cable_angle_deg	41.4097	0.0409	41.3792	41.4561
0.3000	geometric	mean_3d_error_m	0.3841	0.0008	0.3833	0.3849
0.3000	geometric	rms_3d_error_m	0.4006	0.0007	0.3999	0.4013
0.3000	geometric	mean_lateral_swing_m	0.5853	0.0004	0.5849	0.5857
0.3000	geometric	mean_cable_angle_deg	36.0678	0.0311	36.0347	36.0965

Plots

- speed_sweep_tracking_swing.png
- speed_sweep_improvement.png

Interpretation

This is a controlled screening sweep, not a final statistical campaign. It reuses the same validation gates as the repeatability phase and keeps all valid trials in the aggregate.

The main finding is speed dependence. At $\omega=0.15$ rad/s, the PX4 baseline tracks better than the geometric controller and both controllers produce nearly identical swing. At $\omega=0.20$ rad/s, the PX4 baseline still tracks better, but the geometric controller already reduces payload swing. At $\omega=0.25$ rad/s and $\omega=0.30$ rad/s, the geometric controller improves both trajectory tracking and payload swing.

The crossover behavior suggests that the current geometric-controller gains are better suited to moderate and high trajectory rates than to slow Figure-8 motion.

Slow-speed tuning should therefore be treated separately from high-speed payload-swing suppression.

Trajectory Aggressiveness vs. Controller Bandwidth

To address whether the tracking degradation is driven by trajectory aggressiveness or controller bandwidth, we analyze the theoretical peak kinematics of the commanded Figure-8 (amplitude $A = 5.0\text{ m}$):

- **Maximum Velocity:** $v_{\text{max}} = \sqrt{5} A \omega$
- **Maximum Acceleration:** $a_{\text{max}} = 4 A \omega^2$

ω (rad/s)	v_{max} (m/s)	a_{max} (m/s ²)
0.15	1.68	0.45
0.20	2.24	0.80
0.25	2.80	1.25
0.30	3.35	1.80

Interpretation

- Low-Speed Regime ($\omega = 0.15$ rad/s):** At this frequency, kinematic demands are very low ($a_{\text{max}} = 0.45\text{ m/s}^2$). Because the PX4 baseline outperforms the geometric controller here, the geometric controller's degradation is **not** due to trajectory aggressiveness. Instead, it indicates a **tuning/bandwidth issue**—the geometric gains are likely tuned too stiffly for slow, quasi-static reference tracking, resulting in over-correction.
- High-Speed Regime ($\omega \geq 0.25$ rad/s):** As frequency increases, acceleration demands scale quadratically, reaching 1.80 m/s^2 at $\omega = 0.30$. In this regime, tracking errors increase for *both* controllers. This universal degradation points to **trajectory aggressiveness** pushing the physical actuation limits of the

vehicle. However, the geometric controller maintains better bandwidth than the baseline under these high dynamic loads, tracking the aggressive reference significantly better and suppressing payload swing.

Conclusion: High-speed tracking degradation is fundamentally tied to trajectory aggressiveness ($a_{\max}$ scaling). Conversely, the geometric controller's underperformance at low speeds is a bandwidth/tuning artifact, suggesting that gain-scheduling may be required for optimal flight across all trajectory rates.